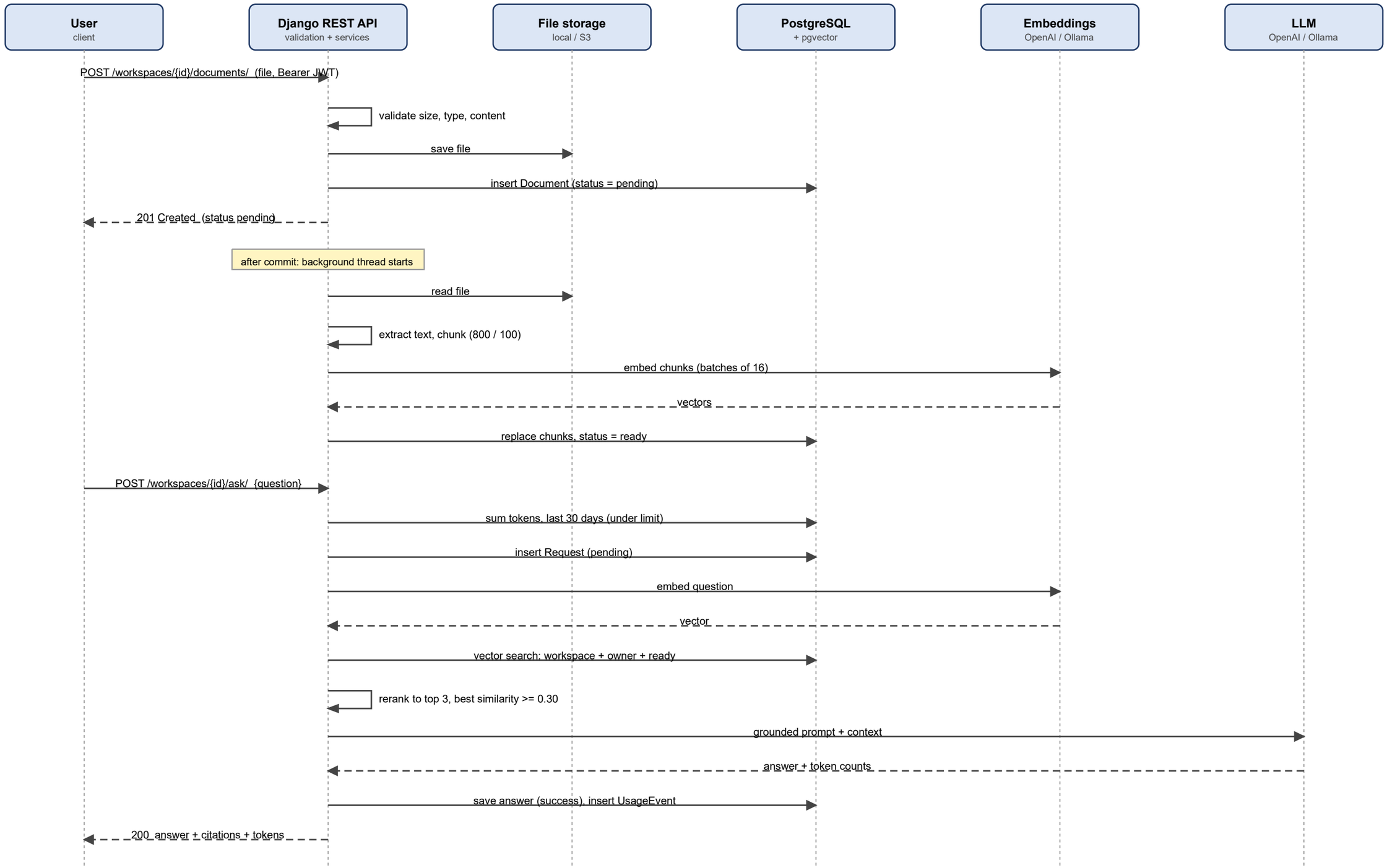
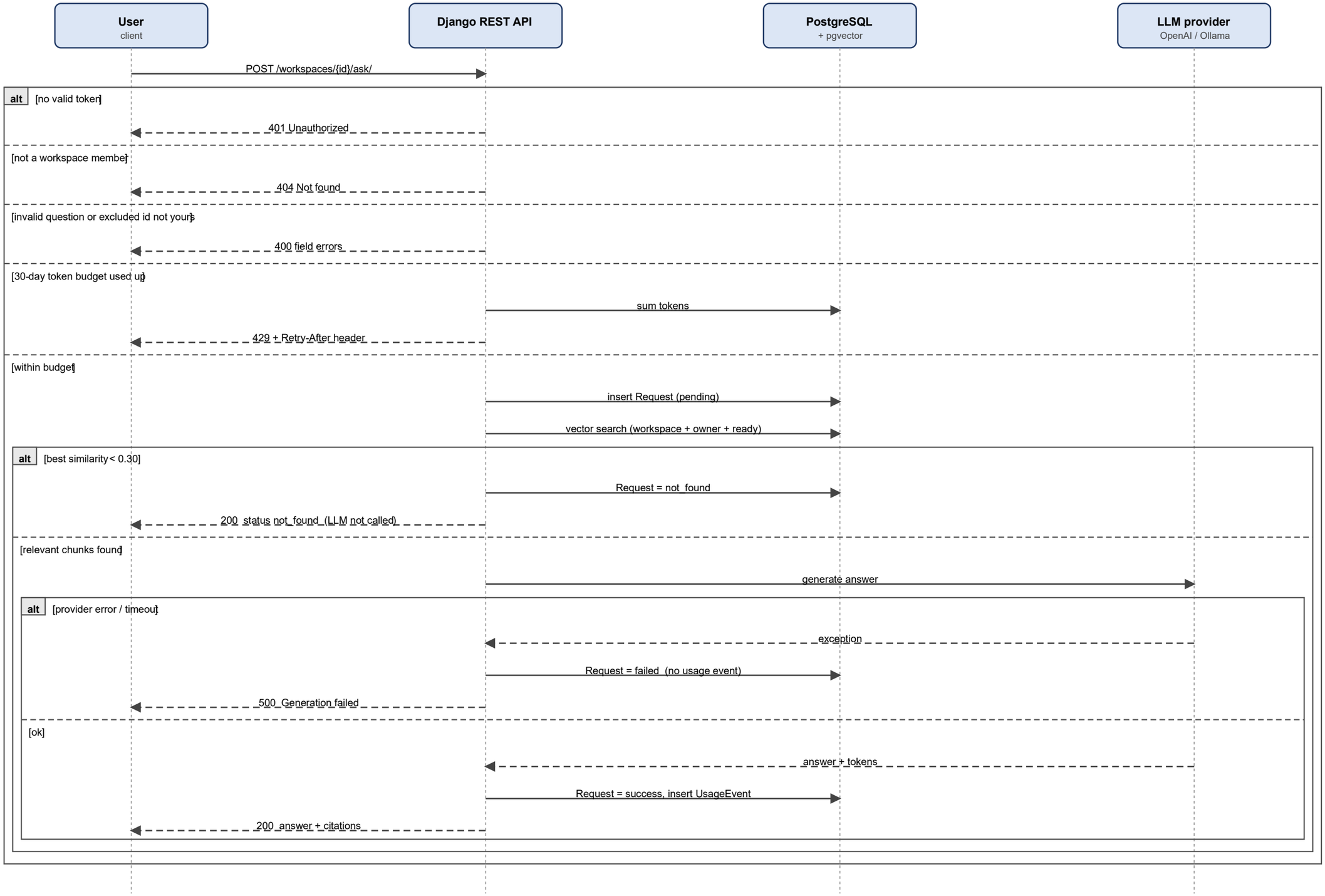
04 - API Workflows: AI SaaS Workspace

Sequence diagrams for the main journeys. Editable Lucidchart URL: **Not made**. Route-by-route detail (request/response examples, validation, errors) is in **04_api-spec.md**.

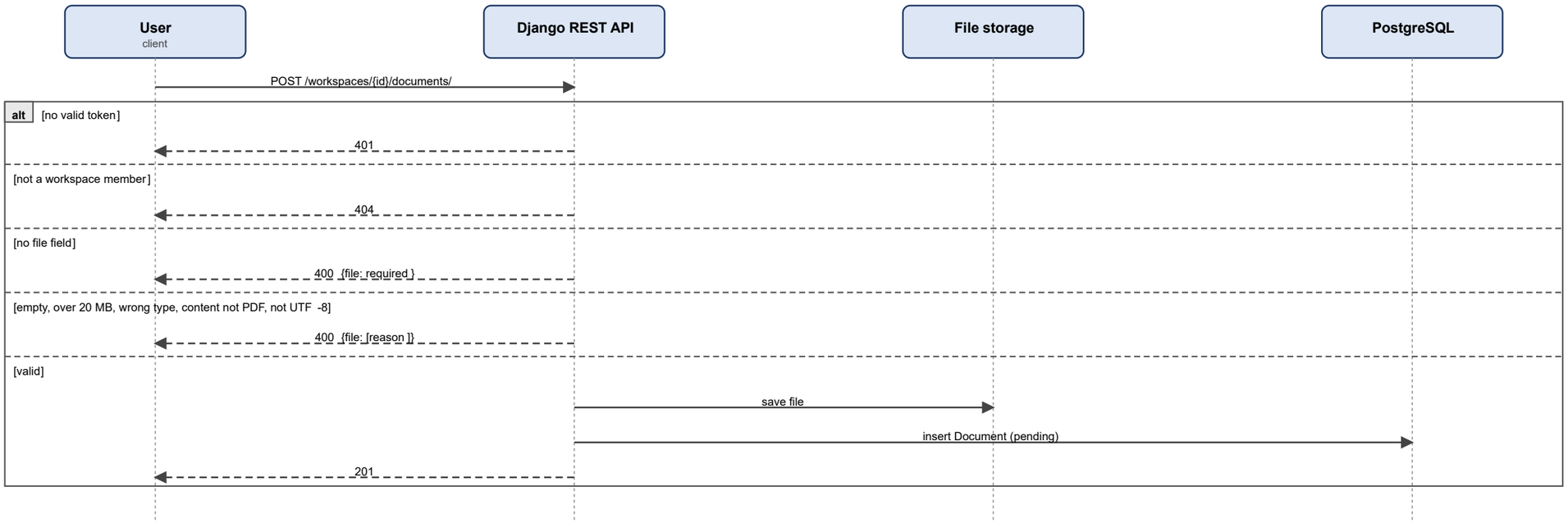
Flow A - Successful journey: upload a document, background ingestion, ask a question



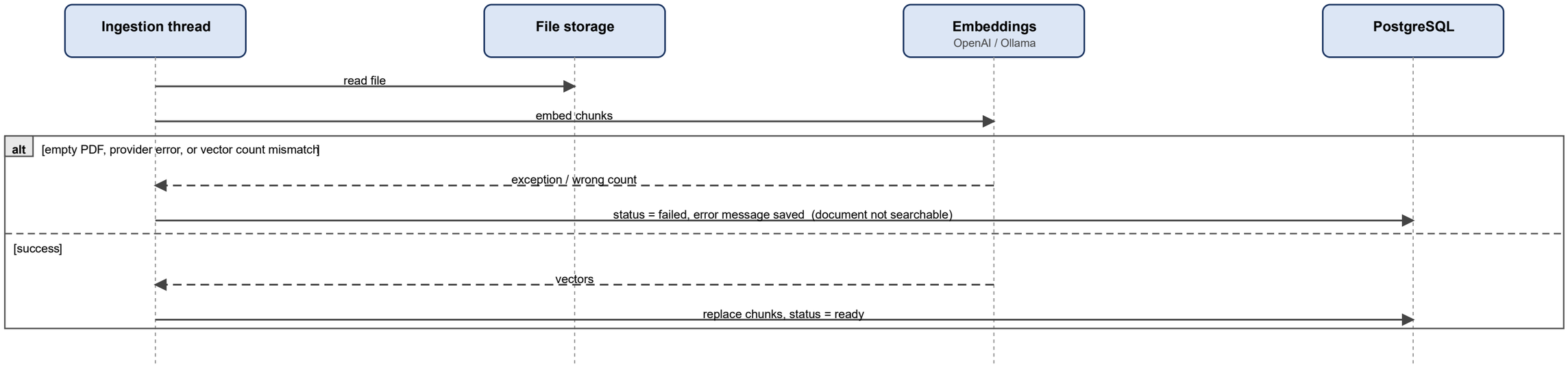
Flow B - Failure journey: asking a question (authentication, membership, validation, usage limit, no answer, provider failure)



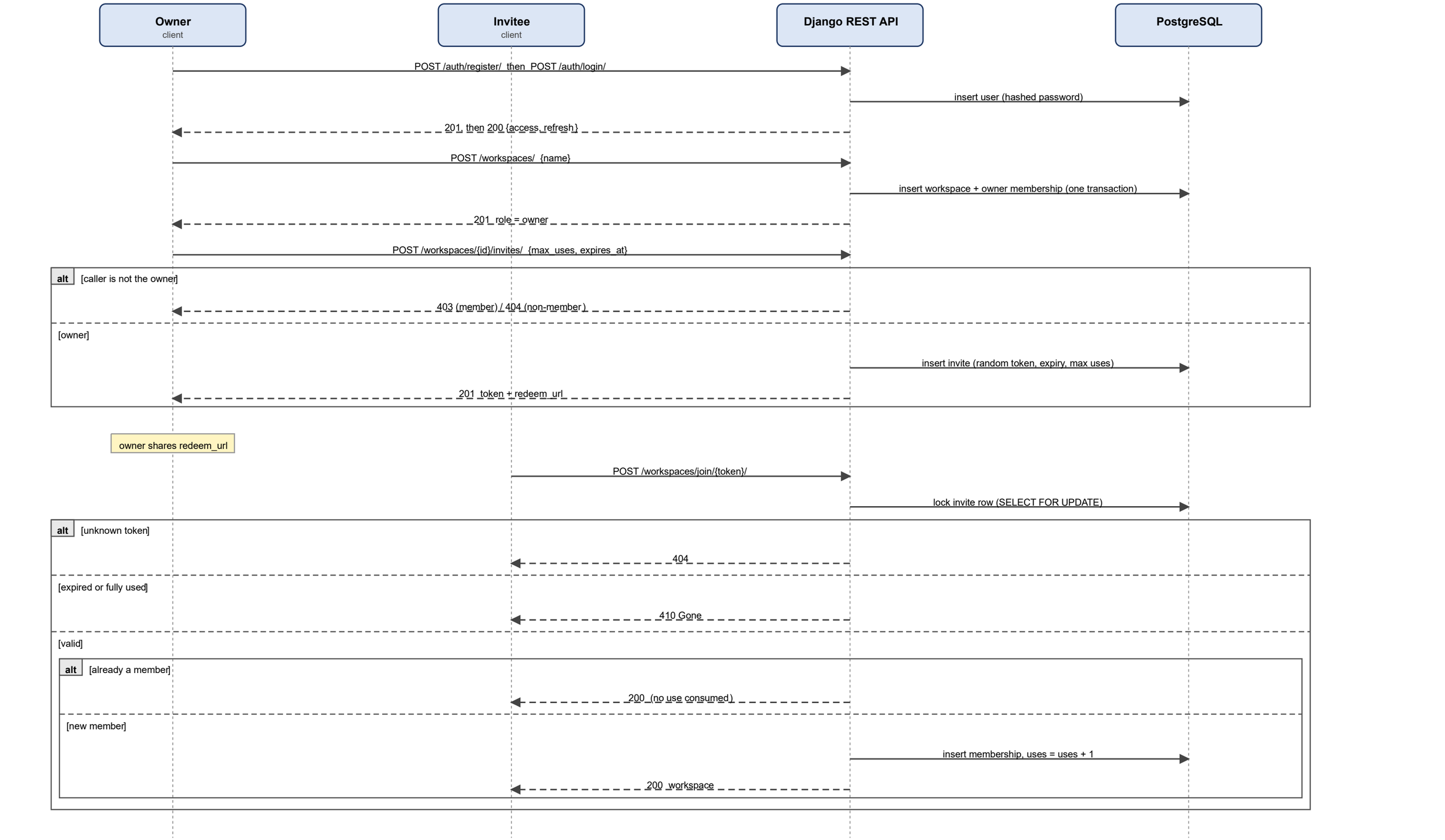
Flow B2 - Failure journey: upload rejected



Flow B3 - Failure journey: ingestion fails after upload



Flow C - Supporting journey: register, create workspace, invite and join (including failures)



Route summary

Method	Path (under /api/v1/)	Auth	Success	Main errors

POST	auth/register/	public	201	400
POST	auth/login/	public (10/min)	200 tokens	401, 429
POST	auth/refresh/	public	200 new tokens (rotated)	401
GET	auth/me/	JWT	200	401
POST	auth/logout/	JWT	204	400, 401
GET / POST	workspaces/	JWT	200 / 201	400, 401
GET	workspaces/{id}/	member	200	404
POST	workspaces/{id}/invites/	owner	201	400, 403, 404
POST	workspaces/join/{token}/	JWT	200	404, 410
GET / POST	workspaces/{id}/documents/	member	200 / 201	400, 404
DELETE	workspaces/{id}/documents/{doc_id}/	uploader	204	403, 404
POST	workspaces/{id}/ask/	member	200 (success or not_found)	400, 404, 429, 500
GET	workspaces/{id}/requests/	member (own)	200	404
GET	workspaces/{id}/requests/{request_id}/	member (own)	200 with citations	404
GET	usage/	JWT	200	401

Idempotency and retry behaviour

Operation	Safe to retry?	Why
Invite redemption	Yes, while the invite is valid	An existing member gets 200 and no extra use is consumed; the invite row is locked during redemption
Document ingestion	Yes	Old chunks are deleted and new ones inserted in one transaction
Upload	No	Each upload creates a new document; clients should not retry blindly
Ask	No	Each call creates a request and consumes tokens if answered; a 500 should be shown to the user, not auto-retried
Logout	Second call fails with 400	The refresh token is already blacklisted
Delete document	Second call returns 404	The document no longer exists